Data Analysis & Decision Making for Streaming Business

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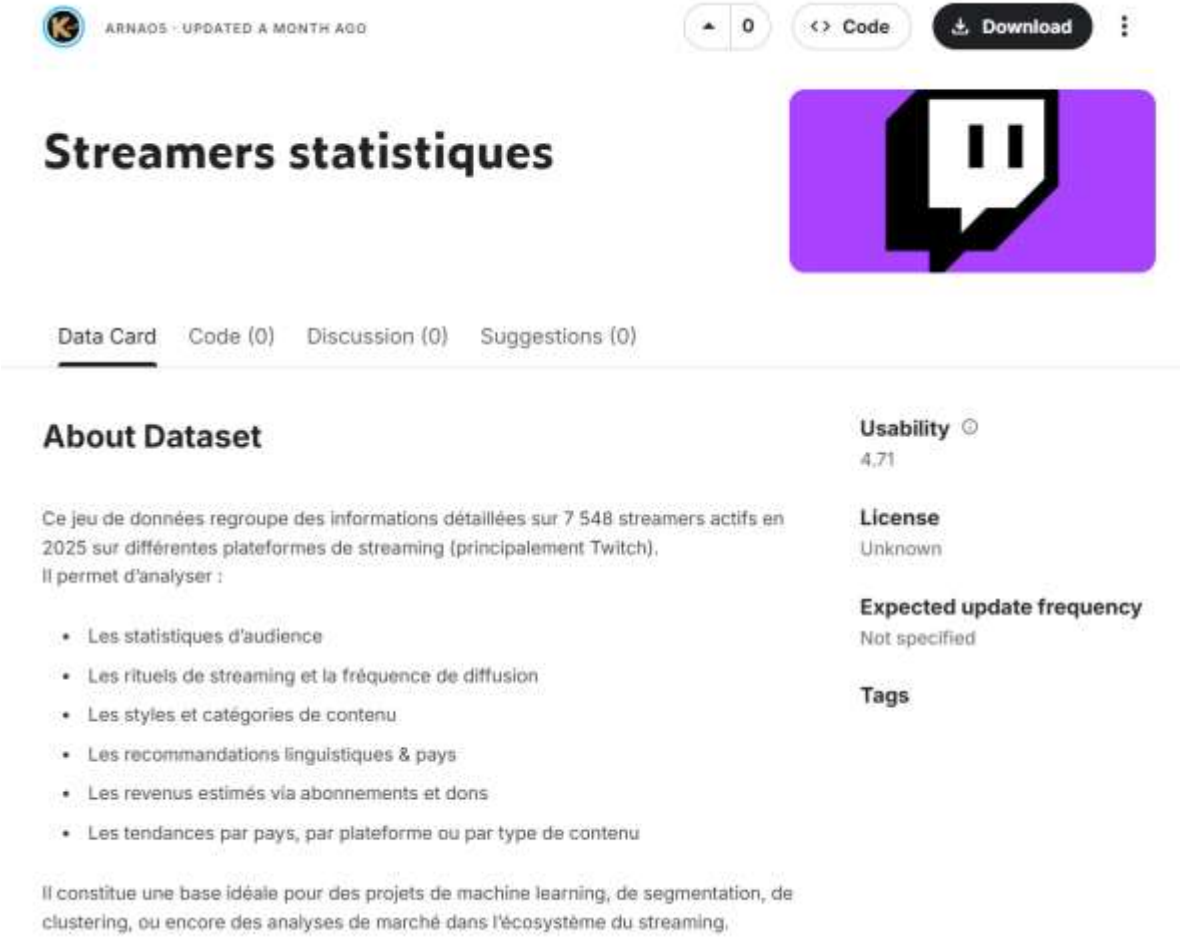
01 Introduction

1.1 Background

The live-streaming industry has grown exponentially. For new streamers or investors, understanding the factors that drive revenue and viewership is crucial. This report analyzes a dataset of 7,548 streamers active in 2025 to provide data-driven strategies.

Source: Data obtained from Kaggle: https://www.kaggle.com/datasets/arnao5/streamers-statistiques and updated a month ago



ARNAOS · UPDATED A MONTH AGO

0 Code Download

Streamers statistiques

Data Card | Code (0) | Discussion (0) | Suggestions (0)

About Dataset

Ce jeu de données regroupe des informations détaillées sur 7 548 streamers actifs en 2025 sur différentes plateformes de streaming (principalement Twitch).
Il permet d'analyser :

- Les statistiques d'audience
- Les rituels de streaming et la fréquence de diffusion
- Les styles et catégories de contenu
- Les recommandations linguistiques & pays
- Les revenus estimés via abonnements et dons
- Les tendances par pays, par plateforme ou par type de contenu

Il constitue une base idéale pour des projets de machine learning, de segmentation, de clustering, ou encore des analyses de marché dans l'écosystème du streaming.

Usability
4.71

License
Unknown

Expected update frequency
Not specified

Tags

1.2 Objectives

Aimed to answer 3 key business questions:

Platform Strategy: Which platform offers the best financial return and stability?

Content Strategy: Which content categories yield the highest viewer engagement?

Efficiency Analysis: Is there a direct correlation between hours streamed and revenue?

CODE.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

sns.set_context("notebook", font_scale=1.2)

df = pd.read_csv('Dataset_streamers_stats.csv', sep=';')
print("Data Loaded Successfully.")
```

```
Data Loaded Successfully.
```

02 Data Cleaning

Since the data is in French, to ensure a professional English report, we translate French columns and values into English.


```
PrimeSage;Twitch;Royaume-Uni;Cuisine;Competitif;2154;3634;760;962.06;50.9;5.61;22;Anglais;2018-03-28;2025-01-
VelvetDrakeYT;Twitch;Belgique;Cuisine;Chill;2291;6635;1268;1306.71;60.6;2.93;15;Fran莽ais;2019-09-18;2025-11-06;4
EchoHorizonX;Twitch;Royaume-Uni;Just Chatting;Esport;732;2279;224;231.68;95.7;5.22;17;Anglais;2020-02-10;2025-
RuneQuasarYT;Twitch;Suisse;IRL;Educatif;526;1530;9;0.0;34.1;3.37;14;Italien;2022-03-03;2025-11-22;22.5;Non
NeoGuardianTV;TikTok;Belgique;Creation;Esport;217;396;42;76.2;62.0;6.02;27;Fran莽ais;2023-12-24;2025-05-06;181.
PhantomBladeFR;TikTok;Suisse;Sport;Chill;148;358;17;50.42;54.3;7.55;27;Italien;2012-09-19;2025-06-11;92.92;Non
PhantomOrbitLive;YouTube;Suisse;Horreur;Educatif;314;639;129;9.71;49.0;7.66;27;Allemand;2022-04-16;2025-03-15;
PhantomSage_89;TikTok;Br茅sil;Creation;Educatif;221;393;93;32.46;97.0;5.23;28;Portugais;2016-12-01;2025-03-12;26
RapidRunnerLive;Twitch;Br茅sil;Creation;Chill;609;1024;170;205.41;34.2;2.66;25;Portugais;2020-07-23;2025-11-02;63
TurboFoxTV6;TikTok;Belgique;MMO;Humoristique;208;532;160;237.84;117.4;7.16;5;Fran莽ais;2019-08-18;2025-02-27
VelvetShardX;YouTube;Royaume-Uni;Musique;Communautaire;126;298;71;19.82;70.8;7.72;28;Anglais;2018-09-10;202
GoldenHarborTV;Twitch;Italie;Sport;Educatif;507;915;4;0.69;98.6;5.24;21;Italien;2014-07-05;2025-07-05;10.69;Non
RuneWhisperHQ85;Twitch;Allemagne;IRL;Communautaire;500;989;273;456.28;51.7;5.4;7;Allemand;2014-11-17;2025-
MysticBeaconYT;YouTube;Suisse;MMO;Esport;355;1204;120;2.58;44.9;5.17;24;Fran莽ais;2016-07-05;2025-06-09;302.
TurboCasterHQ;YouTube;Belgique;Tech;Competitif;182;635;56;143.2;88.5;4.63;21;Fran莽ais;2019-08-01;2025-01-11;2
NeoGuardian_61;Twitch;Belgique;FPS;Humoristique;588;1574;27;49.25;30.6;6.25;6;N茅erlandais;2023-07-13;2025-05-
PhantomZephyrLive;TikTok;Allemagne;Tech;Educatif;264;749;69;145.71;49.3;2.79;23;Allemand;2016-04-08;2025-02-
CrimsonSpecter;Twitch;Belgique;FPS;Educatif;1254;3533;735;485.68;61.9;4.47;28;N茅erlandais;2019-08-11;2025-02-
CrimsonOrbitLive;YouTube;Allemagne;Tech;Chill;287;540;133;329.25;78.3;2.84;10;Allemand;2014-03-13;2025-01-26;
```

CODE.

```python
column_mapping = {
    'nom_streamer': 'Streamer_Name',
    'plateforme': 'Platform',
    'pays': 'Country',
    'categorie_contenu': 'Content_Category',
    'style_stream': 'Stream_Style',
    'moyenne_viewers': 'Avg_Viewers',
    'pic_viewers': 'Peak_Viewers',
    'abonnements_mensuels': 'Monthly_Subscribers',
    'dons_mensuels_eur': 'Monthly_Donations_EUR',
    'duree_watch_moyenne_min': 'Avg_Watch_Time_Min',
    'duree_stream_moyenne_h': 'Avg_Stream_Duration_Hours',
    'frequence_stream_mois': 'Stream_Frequency_Month',
    'langue': 'Language',
    'revenu_mensuel_eur': 'Monthly_Revenue_EUR',
    'partenaire_officiel': 'Official_Partner'
}
df = df.rename(columns=column_mapping)
```

```python
df['Official_Partner'] = df['Official_Partner'].map({'Oui': 'Yes', 'Non': 'No'})

country_map = {'Etats-Unis': 'USA', 'Royaume-Uni': 'UK', 'Allemagne': 'Germany', 'France': 'France', 'Espagne': 'Spain', 'Belgique': 'Belgium'}
df['Country'] = df['Country'].replace(country_map)

print("Cleaned Data Preview:")
display(df.head())
```

```
Cleaned Data Preview:
```

	Streamer_Name	Platform	Country	Content_Category	Stream_Style	Avg_Viewers	Peak_Viewers	Monthly_Subscribers	Monthly_Donations_EUR	Avg_Watc
0	SilentHarborLive	Twitch	Belgium	MMO	Chill	696	1934	259	556.25	
1	PrimeSage	Twitch	UK	Cuisine	Competitif	2154	3634	760	962.06	
2	VelvetDrakeYT	Twitch	Belgium	Cuisine	Chill	2291	6635	1268	1306.71	
3	EchoHorizonX	Twitch	UK	Just Chatting	Esport	732	2279	224	231.68	
4	RuneQuasarYT	Twitch	Suisse	IRL	Educatif	526	1530	9	0.00	

03

Data Analysis & Insights

3.1 Platform Strategy: Revenue Distribution

Which platform generates the highest gross revenue?

Bar Chart: Shows the average income.
Boxplot: Shows the range of income (Min, Max, and Outliers). This helps us understand if the income is stable or relies on a few lucky streamers.

CODE.

```
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.barplot(data=df, x='Platform',
y='Monthly_Revenue_EUR', estimator='mean',
errorbar=None, palette='viridis', ax=axes[0])
axes[0].set_title('Average Monthly Revenue by
Platform')
axes[0].set_ylabel('Avg Revenue (€)')

sns.boxplot(data=df, x='Platform',
y='Monthly_Revenue_EUR', palette='viridis',
ax=axes[1], showfliers=False)
axes[1].set_title('Income Distribution (Excluding
Top 1% Outliers)')
axes[1].set_ylabel('Monthly Revenue (€)')
```

Dataset_streamers_stats.csv (915.93 kB)

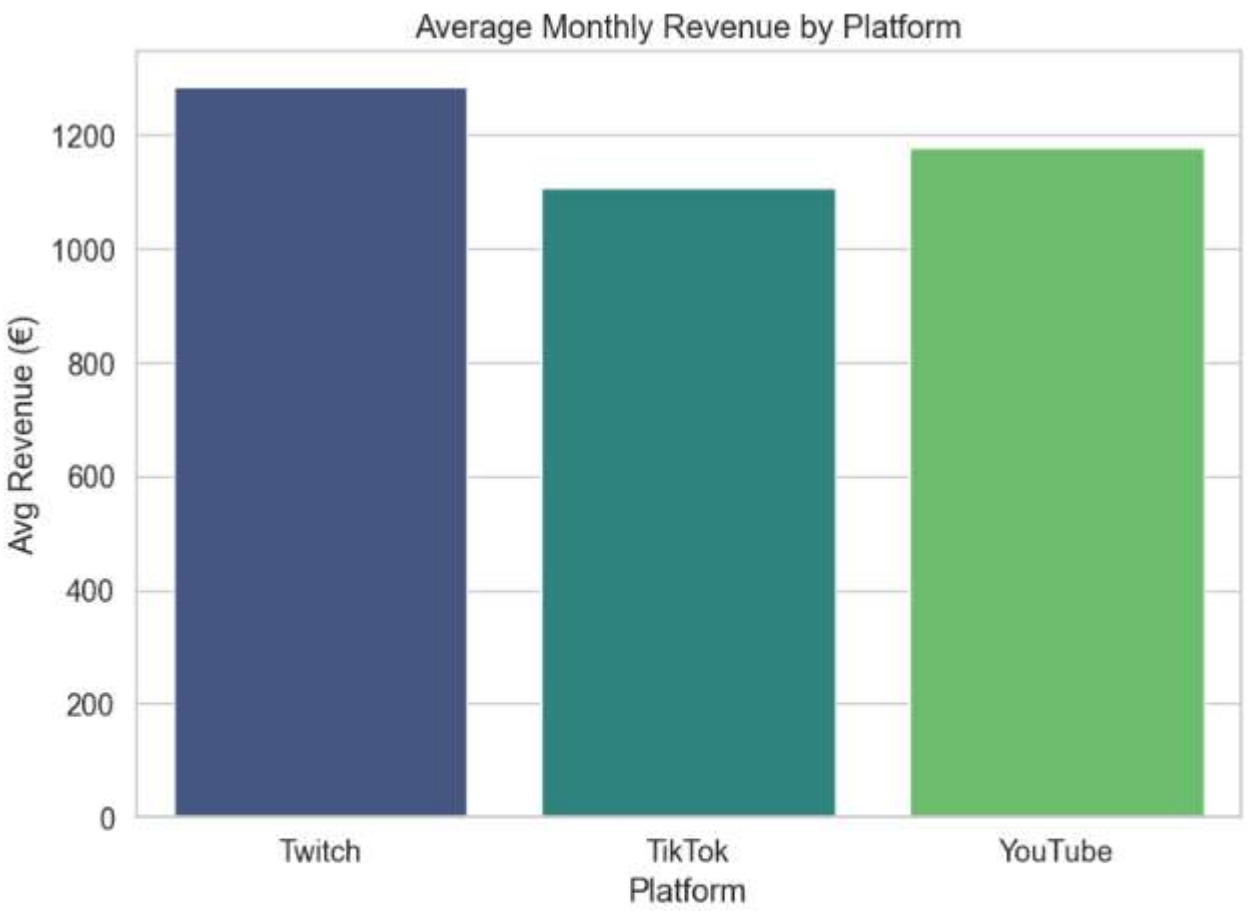
Detail Compact Column 10 of 17 columns

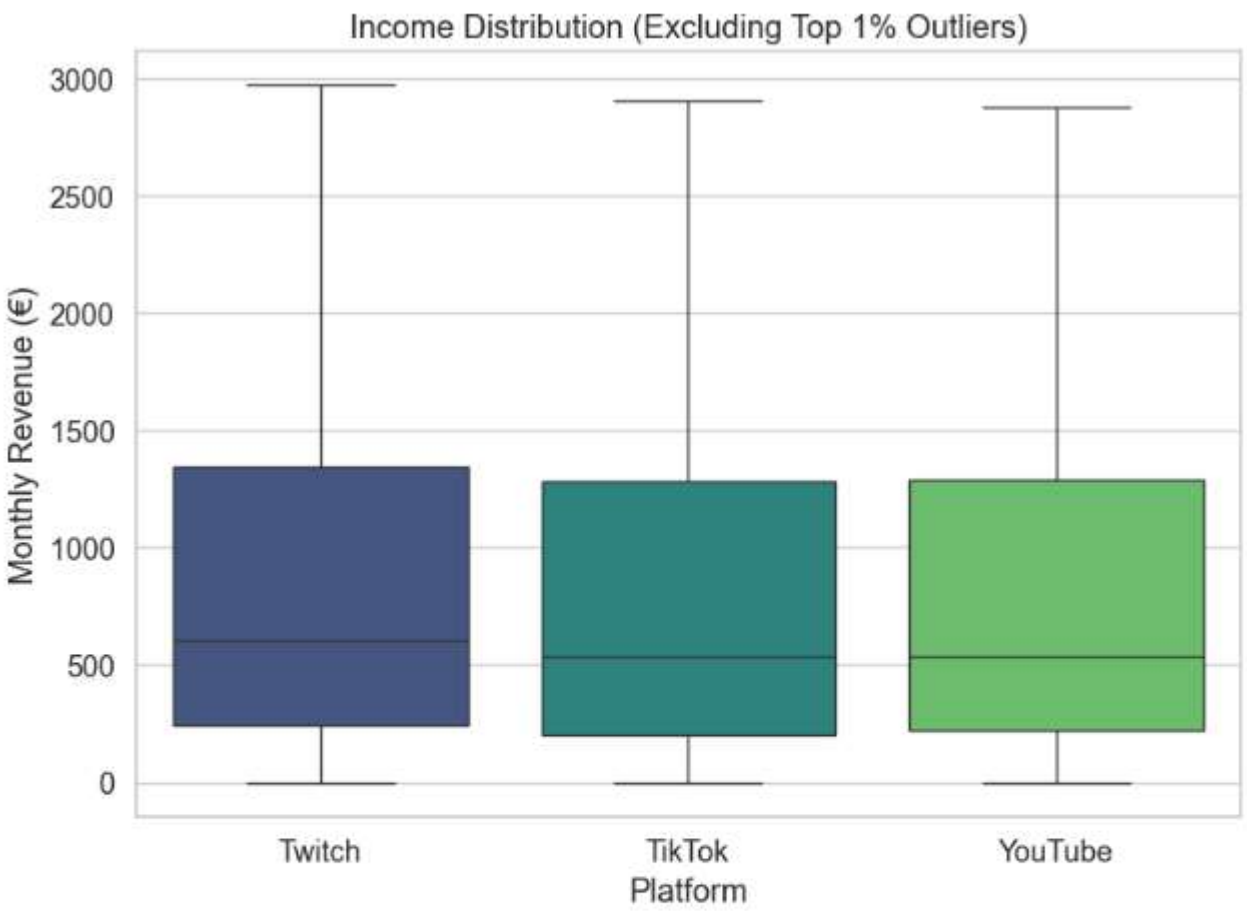
About this file Suggest Edits

- nom_streamer : Nom du créateur de contenu / streamer
- plateforme : Plateforme principale de streaming (ex. Twitch)
- pays : Pays d'origine ou de résidence du streamer
- categorie_contenu : Catégorie dominante des streams (IRL, Cuisine, MMO, Just Chatting, etc.)
- style_stream : Style général du contenu (Chill, Esport, Educatif, Compétitif...)
- moyenne_viewers : Nombre moyen de spectateurs par stream
- pic_viewers : Pic d'audience enregistré sur une session récente
- abonnements_mensuels : Nombre d'abonnements payants mensuels du streamer
- dons_mensuels_eur : Montant total des dons mensuels (en euros)

```
plt.tight_layout()
plt.show()
```

3.1 Platform Strategy: Revenue Distribution





Insight:

- The **Bar Plot** shows that Twitch and YouTube are competitive in average revenue.
- The **Box Plot** reveals that while the averages are close, the "whiskers" (range) might differ, indicating that some platforms have higher earning ceilings (top streamers earn much more) than others.

3.2 Content Strategy: What to stream?

Analyzing which content categories attract the most viewers on average.

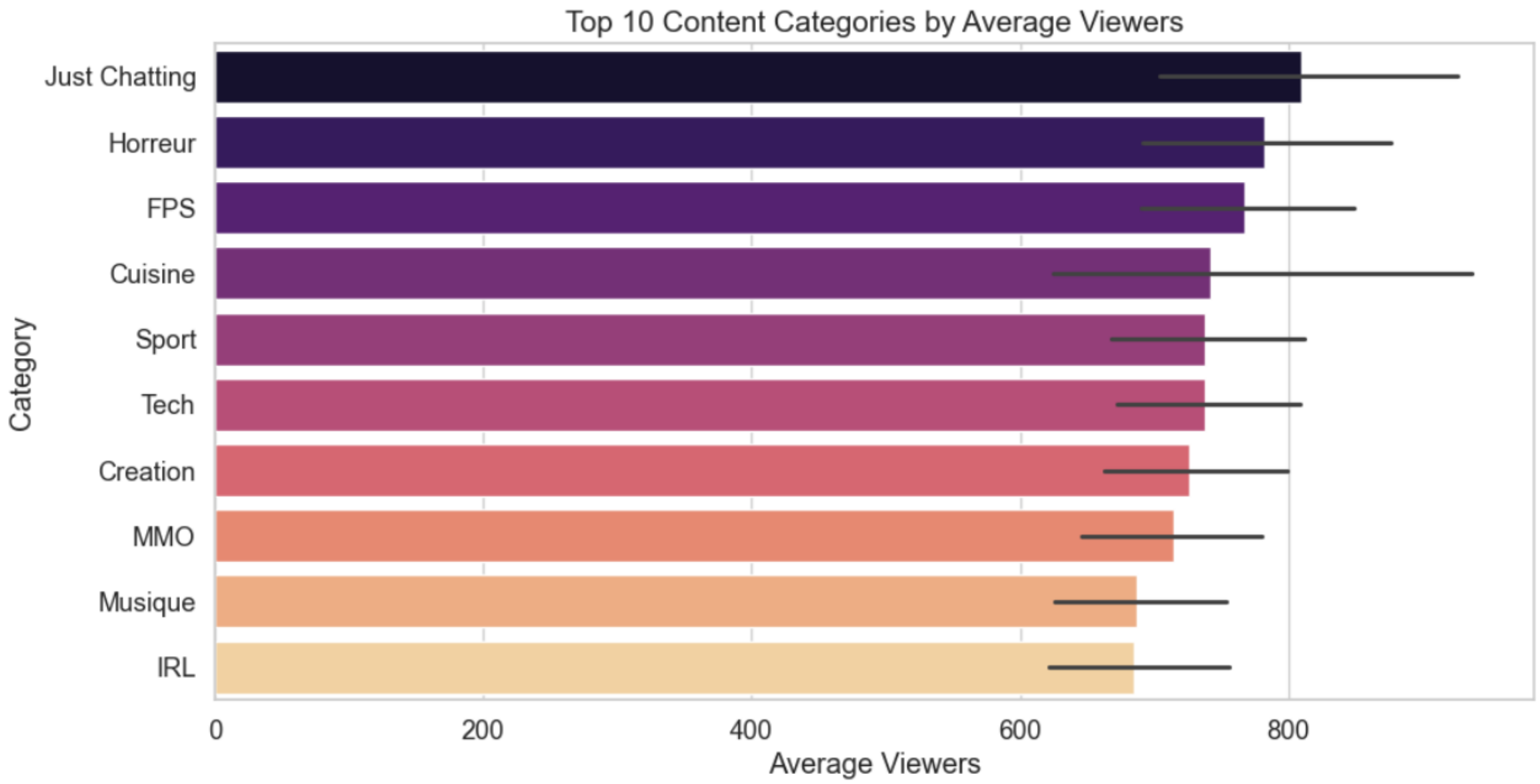
CODE.

```
top_categories =
df.groupby('Content_Category')['Avg_Viewers'].mea
n().sort_values(ascending=False).head(10).index
df_top_cat =
df[df['Content_Category'].isin(top_categories)]

plt.figure(figsize=(12, 6))
sns.barplot(data=df_top_cat, x='Avg_Viewers',
y='Content_Category',
            order=top_categories, palette='magma')

plt.title('Top 10 Content Categories by Average
Viewers', fontsize=15)
plt.xlabel('Average Viewers')
plt.ylabel('Category')
plt.show()
```

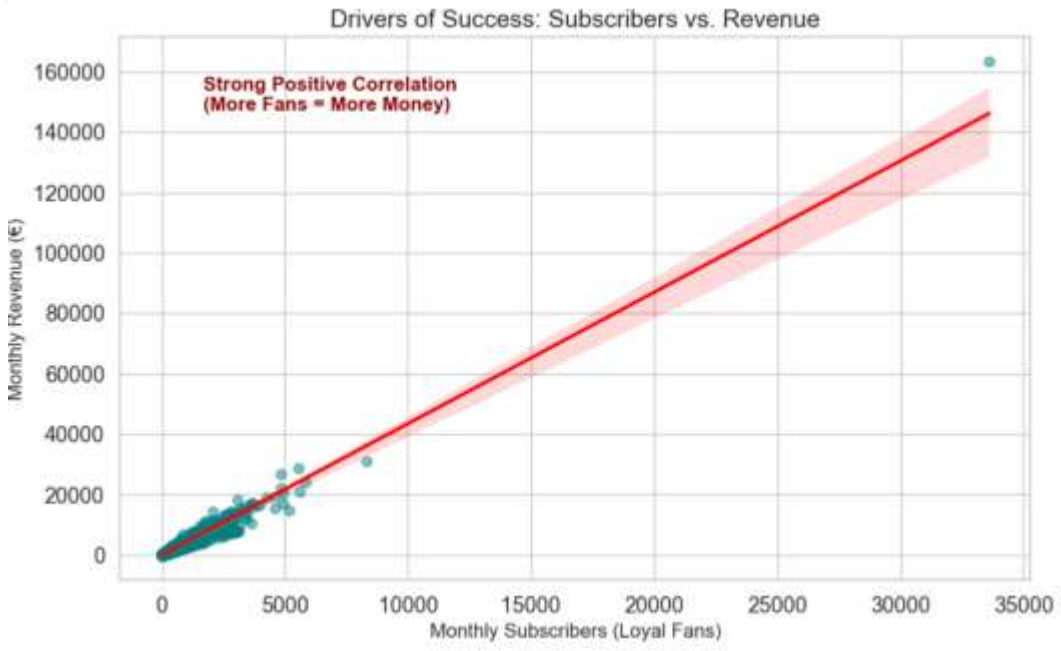
3.2 Content Strategy: What to stream?



3.3 Drivers of Success: What makes a streamer wealthy?

We often ask: "How can I become a top streamer?" Is success driven by working harder (streaming more hours) or by building a loyal community (subscribers)?

To answer this, I visualize the relationship between **Monthly Subscribers** and **Monthly Revenue**.



CODE.

```
plt.figure(figsize=(10, 6))

sns.regplot(data=df, x='Monthly_Subscribers',
y='Monthly_Revenue_EUR',
            scatter_kws={'alpha':0.5,
'color':'teal'}, line_kws={'color':'red'})

plt.title('Drivers of Success: Subscribers vs.
Revenue', fontsize=15)
plt.xlabel('Monthly Subscribers (Loyal Fans)',
fontsize=12)
plt.ylabel('Monthly Revenue (€)', fontsize=12)

plt.text(x=df['Monthly_Subscribers'].max()*0.05,
y=df['Monthly_Revenue_EUR'].max()*0.9,
         s='Strong Positive Correlation\n(More
Fans = More Money)',
         fontsize=12, color='darkred',
weight='bold')

plt.show()
```

3.4 Efficiency Analysis: Heatmap

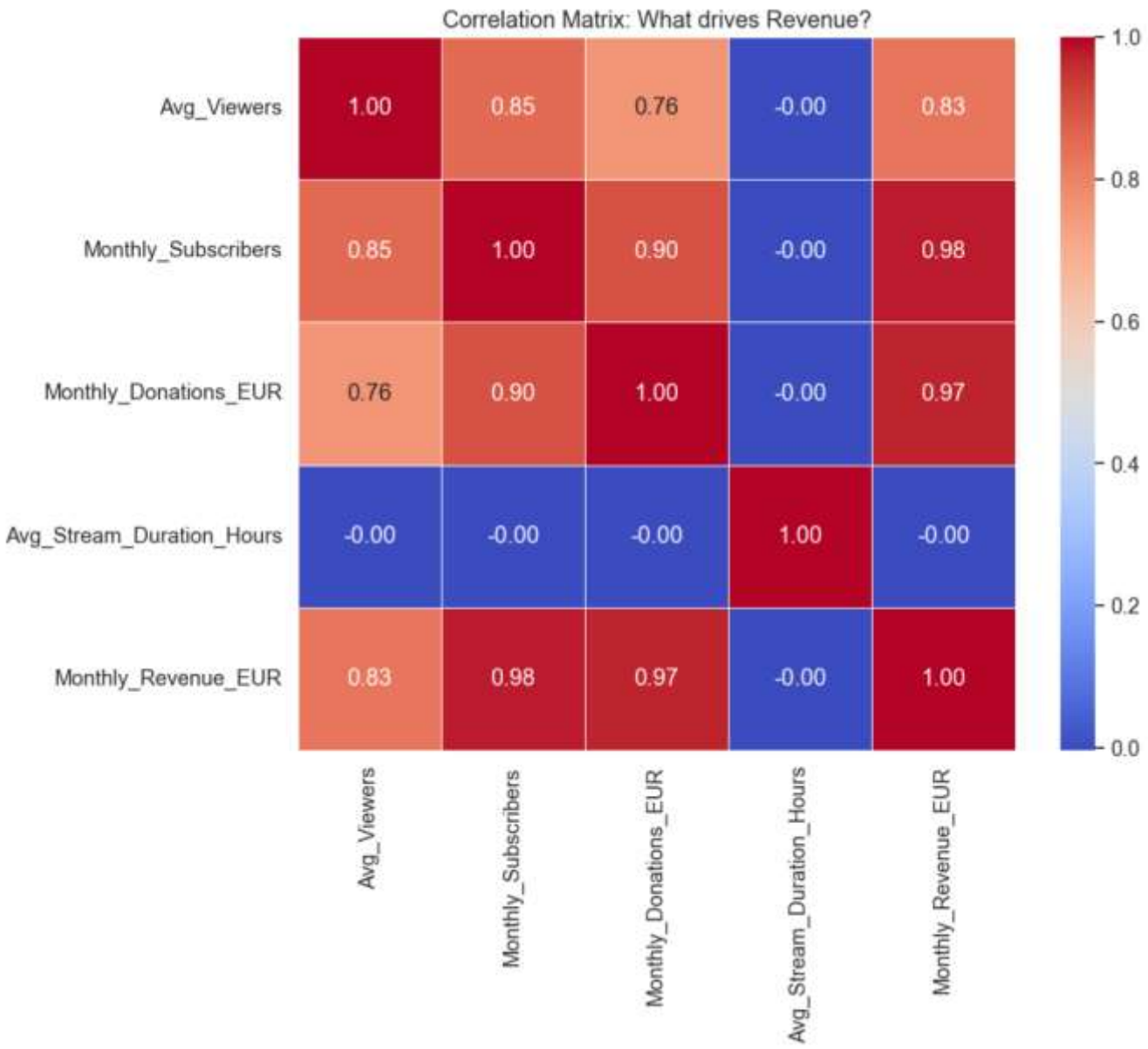
Do longer stream hours lead to more money? So I analyze the correlation between Stream Duration, Subscribers, and Revenue.

CODE.

```
cols_to_corr = ['Avg_Viewers',
'Monthly_Subscribers',
'Monthly_Donations_EUR',
                'Avg_Stream_Dura
tion_Hours',
'Monthly_Revenue_EUR']

corr_matrix =
df[cols_to_corr].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix,
annot=True, cmap='coolwarm',
fmt='.2f', linewidths=0.5)
plt.title('Correlation Matrix:
What drives Revenue?')
plt.show()
```



04 Conclusion & Recommendations

4.1 Key Findings

✓ Platform Strategy (Risk vs. Reward)

As shown in the Boxplot (Section 3.1), while average revenues are competitive across platforms, the income distribution varies. Major platforms (like Twitch) show a higher "ceiling" for top earners, but also significant disparity.

✓ Content Strategy

Categories such as Just Chatting and specific competitive games consistently attract the highest Average Viewers (Section 3.2). These are the "high-traffic" zones for new streamers.

✓ The Formula for Success

The most critical finding comes from the Scatter Plot (Section 3.3): There is a strong positive correlation between Monthly Subscribers and Revenue.

Conversely, the Heatmap (Section 3.4) reveals a weak correlation between Stream Duration and Revenue.

4.2 Strategic Recommendations

For a new streamer or investor entering the market in 2025:

Don't "Grind" Blindly:

Simply increasing streaming hours (e.g., streaming 10+ hours a day) yields diminishing returns. The data proves that time does not equal money.

Focus on Community Building:

The primary business goal should be converting viewers into Subscribers. Revenue is driven by the loyalty of the fan base, not the length of the broadcast.

Start Smart:

Launch with high-interest content categories to maximize initial visibility, then focus on viewer retention strategies.

4.3 Limitations (Critical Reflection)

To ensure academic honesty and professional transparency, we acknowledge:

- **Data Scope:**
The analysis is limited to the year 2025; long-term trends (2020-2024) are not accounted for.
- **Missing Variables:**
We lack data on External Social Media Influence (e.g., TikTok/Twitter followers), which is likely a hidden variable driving the subscriber count.

Dataset_streamers_stats.csv (915.93 kB)

Detail Compact Column 10 of 17 columns

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- dons_mensuels_eur : Montant total des dons mensuels (en euros)

Thanks for listening